

SAFE-BY-DESIGN SCREENING & GOVERNANCE

Safe*Pro*

A pre-synthesis review layer that ranks protein candidates by developability and flags *biosecurity risk* before any lab work begins.

TEAM

Dalan

PROBLEM STATEMENT

Computational pre-clinical drug development

01

THE PRE-CLINICAL BOTTLENECK

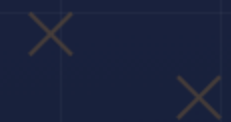
Generating candidates is easy. *Choosing* them isn't.

Generative models now propose novel protein structures far faster than any lab can synthesize and validate them.

- AI generation outpaces lab synthesis capacity
- Manufacturing problems surface late, after resources are spent
- Safety and governance review lags behind the pace of design



SYNTHESIS
CAPACITY

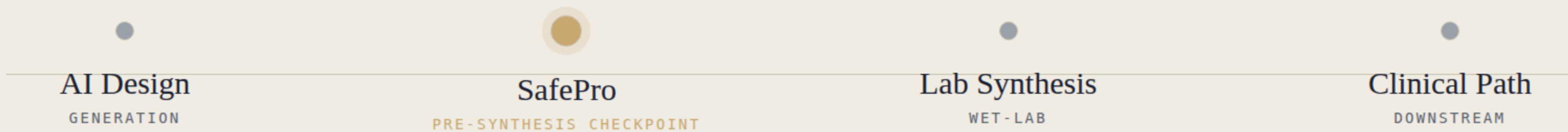


LATE FAILURES

A CHECKPOINT, NOT A BRAKE

SafePro reviews every design *before* it reaches the lab.

It scores each candidate for developability and manufacturability, and screens it for biosecurity and toxicity risk — turning a flood of AI designs into a short, defensible shortlist.



HOW IT WORKS

From sequence to a *defensible* decision.

INPUT

Candidate

Protein sequence and structural data
from upstream AI design models.

SCORING

Two modules

Developability — aggregation & stability.
Biosecurity — toxicity & hazard patterns.

DECISION & AUDIT

PASS

Ready for synthesis

REVIEW

Human-in-the-loop expert escalation

BLOCK

Flagged for severe liabilities

EVERY FLAG IS WRITTEN TO A SECURE AUDIT LOG WITH REVIEWER ANNOTATIONS.

WHY PFIZER

Built for Pfizer's *pipeline* and its values.

PIPELINE

Strengthens Stage 1 discovery

Triages candidates across biologics, vaccines, and oncology programs before they advance toward trials.

MANUFACTURING

De-risks the supply chain early

Filtering out poorly developable designs protects capacity at Pfizer's Global Supply Sites.

VALUES

Responsible AI, by design

Audit logs and human review embed integrity and transparency directly into the decision.

WHAT STANDS IN THE WAY

We know where it gets *hard* — and how we'd answer.

CHALLENGE

Access to robust, normalized training data



MITIGATION

Start with public protein databases, then scale to internal assay data in a pilot phase

CHALLENGE

Managing false positives



MITIGATION

Human-in-the-loop expert review on every flag, so no good candidate is dropped silently

WHAT SUCCESS LOOKS LIKE

Safe, scalable *innovation*.

- Fewer dead-ends late in development
- A safer path for pre-clinical discovery
- Responsible AI built *into* the workflow, not bolted on